

Change the order of following integrals and evaluate (if possible).

1. $\int_0^1 \int_{4y}^4 e^{x^2} dx dy$

$y =$ $x =$

$$\begin{array}{c|c|c|c} y=0 & & y=1 & \\ \hline x=4y & & x=4 & \end{array}$$

$$\int_0^4 \int_{x/4}^1 e^{x^2} dy dx$$

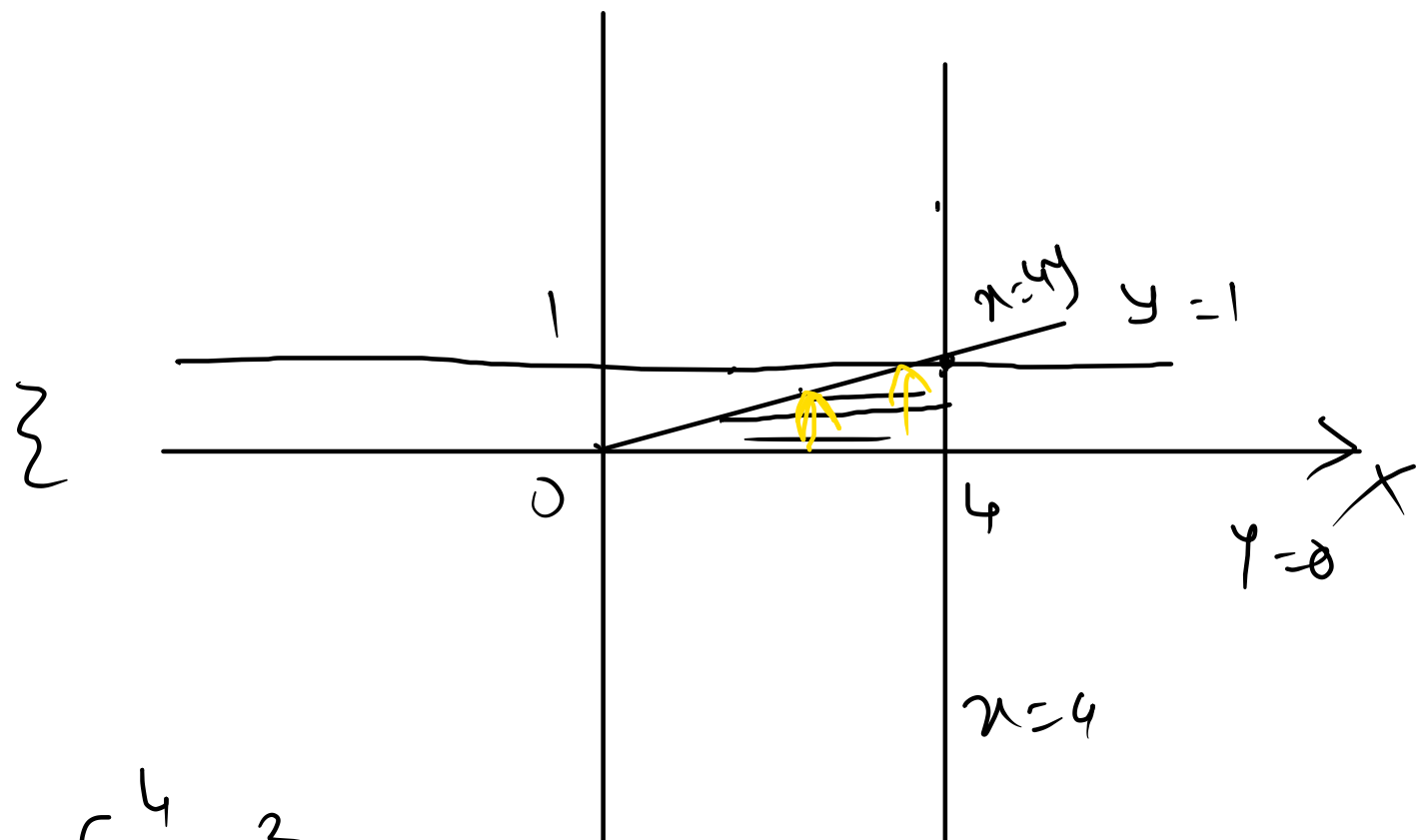
$$x=0 \quad y=0$$

$$= \int_0^4 e^{x^2} (y)_0^{x/4} dx =$$

$$\int_0^4 e^{x^2} \frac{x}{4} dx$$

put $x^2 = t$

$$= \frac{1}{8} (e^{16} - 1)$$



$$2. \int_0^1 \int_0^{\sqrt{1-x^2}} \frac{e^y}{(e^y+1)\sqrt{1-x^2-y^2}} dy dx$$

$$x=0, x=1$$

$$y=0, y=\sqrt{1-x^2}$$

$$y^2 = 1 - x^2$$

$$x^2 + y^2 = 1$$

$$\int_{y=0}^1 \int_{x=0}^{\sqrt{1-y^2}} \frac{e^y}{(e^y+1)\sqrt{1-x^2-y^2}} dx dy$$

$$= \int_0^1 \frac{e^y}{e^y+1} \int_0^a \frac{dx}{\sqrt{a^2-x^2}} dy$$

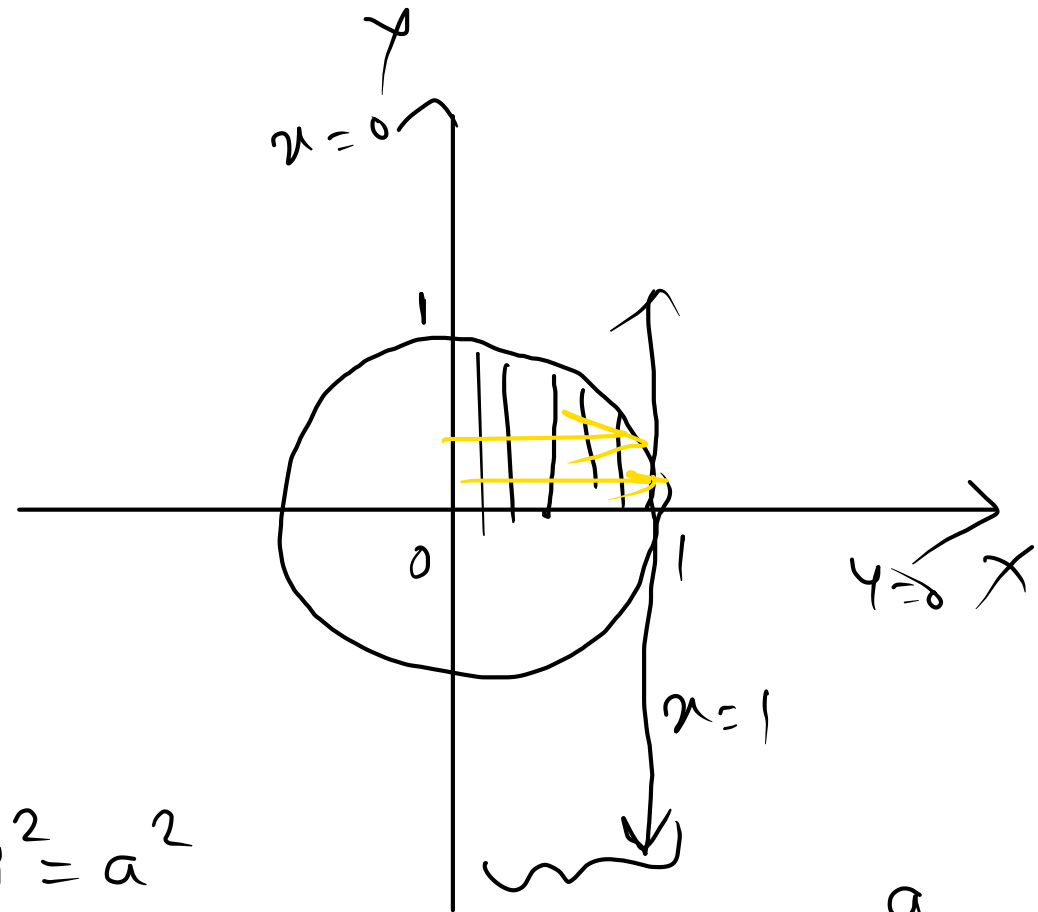
$$1-y^2=a^2$$

$$= \int_0^1 \frac{e^y}{e^y+1} \left[\sin^{-1} \left(\frac{x}{a} \right) \right]_0^a dy$$

$$= \int_0^1 \frac{e^y}{e^y+1} \left(\frac{\pi}{2} \right) dy$$

$$= \frac{\pi}{2} \left[\log(e^y+1) \right]_0^1$$

$$= \frac{\pi}{2} \left[\log(e+1) - \log(2) \right]$$



$$4. \int_0^2 \int_{2-\sqrt{4-y^2}}^{2+\sqrt{4-y^2}} dx dy$$

$y =$ $x =$

$$y = 0$$

$$y = 2$$

$$x = 2 \pm \sqrt{4-y^2}$$

$$(x-2)^2 + y^2 = 4$$

center (2, 0) & rad. 2

$$\int_{x=0}^4 \int_{y=0}^{\sqrt{4-(x-2)^2}} dy dx$$

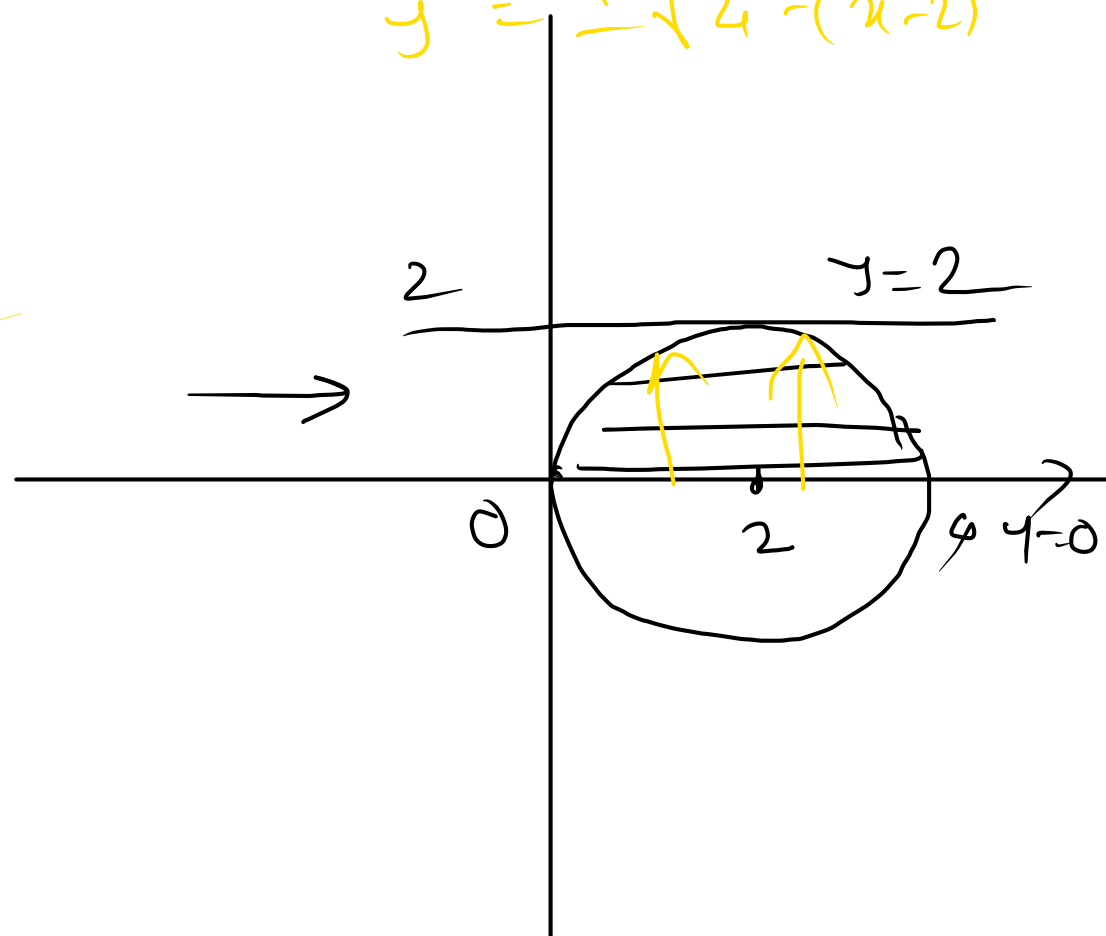
$$\int_0^4 (y) \sqrt{4-(x-2)^2} dx = \int_0^4 \sqrt{4-(x-2)^2} dx$$

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right)$$

$x = x-2$
 $= 2\pi$

$$y^2 = 4 - (x-2)^2$$

$$y = \pm \sqrt{4 - (x-2)^2}$$

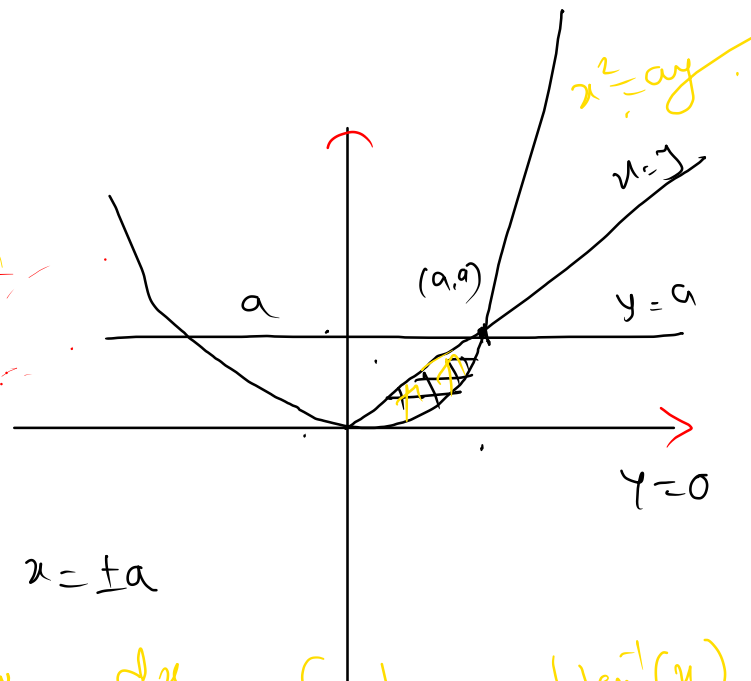


5. $\int_0^a \int_y^{\sqrt{ay}} \frac{x}{x^2+y^2} dx dy$
 $y=x$

$y=0$
 $x=y$
 $y=a$
 $x=\sqrt{ay}$

$x^2=ay$
 $x=\pm\sqrt{ay}$

$y=a$ $x=\pm a$



$\int_0^a \int_{x^2/a}^x \frac{x}{x^2+y^2} dy dx$
 $x=0$ $y=x^2/a$

$= \int_0^a x \int_{x^2/a}^x \frac{1}{y^2+x^2} dy dx$

$\int \frac{1}{x^2+a} = \frac{1}{a} \tan^{-1}\left(\frac{y}{a}\right)$

$= \int_0^a x \left[\tan^{-1}\left(\frac{y}{x}\right) \right]_{x^2/a}^x dx$

$= \int_0^a \left[\frac{\pi}{4} - \tan^{-1}\left(\frac{x}{a}\right) \right] dx$

$= \frac{\pi}{4} (x)_0^a - \int_0^a \tan^{-1}\left(\frac{x}{a}\right) 1 dx$

$= \frac{\pi}{4} a - \left[\tan^{-1}\left(\frac{x}{a}\right) x - \int \frac{x(1)}{1+x^2/a^2} dx \right]_0^a$

$= \frac{a\pi}{4} - \left[x \tan^{-1} \frac{x}{a} - a^2 \left(\frac{1}{2} \int \frac{2x}{a^2+x^2} dx \right) \right]_0^a$

$= \frac{a\pi}{4} - \left[x \tan^{-1} \frac{x}{a} - \frac{a^2}{2} \log(a^2+x^2) \right]_0^a$

$\left\{ \right.$

$= \frac{a}{2} \log 2$

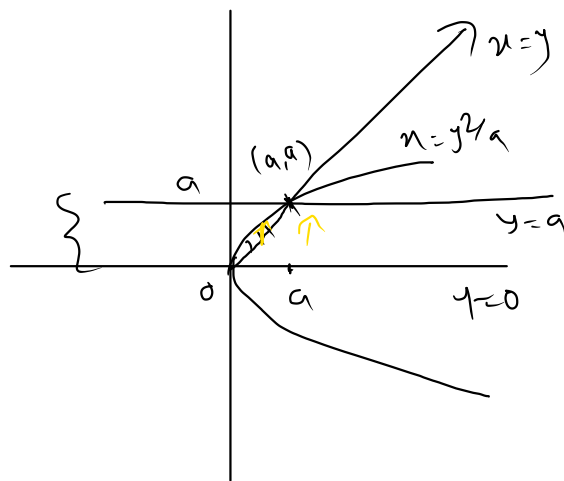
6. $\int_0^a \int_{x=y^2/a}^y \frac{y}{(a-x)\sqrt{ax-y^2}} dx dy$

$y=a$
 $x = a^2/a$

$y=0$
 $x=y^2/a$
 $y=a$
 $x=y$

$y^2 = ax$

$y = \pm \sqrt{ax}$



$\int_{x=0}^a \int_{y=x}^{\sqrt{ax}} \frac{y}{(a-x)\sqrt{ax-y^2}} dy dx$

$= \int_0^a \frac{1}{a-x} \int_{\sqrt{x^2}}^0 \frac{1}{\sqrt{t}} \left(-\frac{dt}{2}\right) dx$

$ax-y^2=t \Rightarrow -2y dy = dt$
 $y dy = -\frac{dt}{2}$

$y \rightarrow \sqrt{ax}$
 $t: ax-x^2 \rightarrow 0$

$t^{-1/2} = \frac{t^{-1/2}}{1/2}$

$= \int_0^a \frac{1}{a-x} \int_0^{ax-x^2} t^{-1/2} \frac{dt}{2} dx$

$= \frac{1}{2} \int_0^a \frac{1}{a-x} \left[\frac{t^{1/2}}{1/2} \right]_0^{ax-x^2} dx = \int_0^a \frac{1}{a-x} (ax-x^2)^{1/2} dx$

$= \int_0^a \frac{1}{a-x} x^{1/2} (a-x)^{1/2} dx = \int_0^a x^{1/2} (a-x)^{-1/2} dx$

$= \int_0^a x^{1/2} a^{-1/2} \left(1-\frac{x}{a}\right)^{-1/2} dx$

$\frac{x}{a} = t \Rightarrow dx = a dt$

$x: 0 \rightarrow a \quad t: 0 \rightarrow 1$

$= \int_0^1 (at)^{1/2} a^{-1/2} (1-t)^{-1/2} a dt$

$= a^{1/2} a^{-1/2} a \int_0^1 t^{1/2} (1-t)^{-1/2} dt$

$= a \beta(3/2, 1/2)$

$= a \frac{\Gamma(3/2) \Gamma(1/2)}{\Gamma(2)} = a \frac{1/2 \sqrt{\pi} \sqrt{\pi}}{1}$

$= a \frac{\pi}{2}$

$\Gamma(1/2) = \sqrt{\pi}$

14. $\int_{x=0}^1 dx \int_{y=1}^{\infty} e^{-y} y^x \log y \, dy$

Note
when limits are

constants
we can change
the order
directly.

$$\int_0^1 \int_1^{\infty} e^{-y} y^x \log y \, dy \, dx$$

$$= \int_{y=1}^{\infty} \int_{x=0}^1 e^{-y} y^x \log y \, dx \, dy$$

$$= \int_1^{\infty} e^{-y} \log y \left[\frac{y^x}{\log y} \right]_0^1 dy$$

$$= \int_1^{\infty} e^{-y} \cancel{\log y} \left[\frac{y-1}{\cancel{\log y}} \right] dy$$

$$= \int_1^{\infty} \frac{(y-1)}{u} \frac{e^{-y}}{v} dy = \left[(y-1) \left(\frac{e^{-y}}{-1} \right) - \int \frac{e^{-y}}{(-1)} (1) dy \right]_1^{\infty}$$

$$= \left[(y-1) \cdot (-e^{-y}) + \frac{e^{-y}}{(-1)} \right]_1^{\infty} = \left[0 - 0 - 0 - \frac{e^{-1}}{-1} \right]$$

$$e^{-\infty} = \frac{1}{e^{\infty}} = 0$$

$$= \frac{1}{e}$$

$$\int a^x dx = \frac{a^x}{\log a}$$

$$\int uv = uv - \int (v u')$$

change the order

$$7. \int_0^a dy \int_0^{a-\sqrt{a^2-y^2}} \frac{xy \log(x+a)}{(x-a)^2} dx$$

$$y = \int_0^a \int_0^{a-\sqrt{a^2-y^2}} \frac{xy \log(x+a)}{(x-a)^2} dx dy$$

$$y=0 \quad y=a$$

$$x=0 \quad x=a-\sqrt{a^2-y^2}$$

$$x-a = -\sqrt{a^2-y^2}$$

$$(x-a)^2 = a^2-y^2$$

$$\Rightarrow (x-a)^2 + y^2 = a^2$$

circle (a,0) rad. a

$$y^2 = a^2 - (x-a)^2$$

$$y = \pm \sqrt{a^2 - (x-a)^2}$$

$$\int_0^a \int_{\sqrt{a^2-(x-a)^2}}^a \frac{xy \log(x+a)}{(x-a)^2} dy dx$$

$$= \int_0^a \frac{x \log(x+a)}{(x-a)^2} \left[\frac{y^2}{2} \right]_{\sqrt{a^2-(x-a)^2}}^a dx$$

$$= \int_0^a \frac{x \log(x+a)}{(x-a)^2} \left[\frac{a^2}{2} - \frac{(a^2 - (x-a)^2)}{2} \right] dx$$

$$= \frac{1}{2} \int_0^a \frac{x \log(x+a)}{(x-a)^2} (x-a)^2 dx$$

$$= \frac{1}{2} \int_0^a \log(x+a) x dx$$

$$= \frac{1}{2} \left[\log(x+a) \frac{x^2}{2} - \int \frac{x^2}{2} \frac{1}{x+a} dx \right]_0^a$$

$$= \frac{1}{2} \left[\log(x+a) \frac{x^2}{2} - \frac{1}{2} \int \left(\frac{x^2 - a^2 + a^2}{x+a} \right) dx \right]_0^a$$

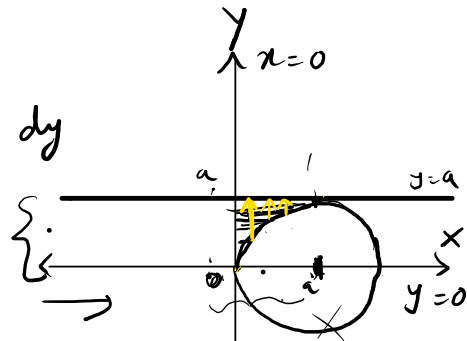
$$= \frac{1}{2} \left[\log(x+a) \frac{x^2}{2} - \frac{1}{2} \int \left[\frac{(x-a)(x+a)}{x+a} + \frac{a^2}{x+a} \right] dx \right]_0^a$$

$$= \frac{1}{2} \left[\frac{x^2}{2} \log(x+a) - \frac{1}{2} \left[\frac{x^2}{2} - ax + a^2 \log(x+a) \right] \right]_0^a$$

$$= \frac{1}{2} \left[\frac{a^2}{2} \log(2a) - \frac{1}{2} \left(\frac{a^2}{2} - a^2 + a^2 \log(2a) \right) - 0 + \frac{a^2}{2} (\log a) \right]$$

$$= \frac{1}{2} \left[\frac{a^2}{2} \log(2a) - \frac{1}{2} \left(-\frac{a^2}{2} \right) - \frac{a^2}{2} \log(2a) + \frac{a^2}{2} (\log a) \right]$$

$$= \frac{1}{2} \left[\frac{a^2}{2} \log(a) + \frac{a^2}{4} \right] //$$



$$\int \int \underline{\underline{1}} dy dx = \underline{\underline{Area}}$$

$$\int_0^5 \int_{2-x}^{2+x} dy dx$$

$x=y$

$$x=0$$

$$y=2-x$$

$$x=5$$

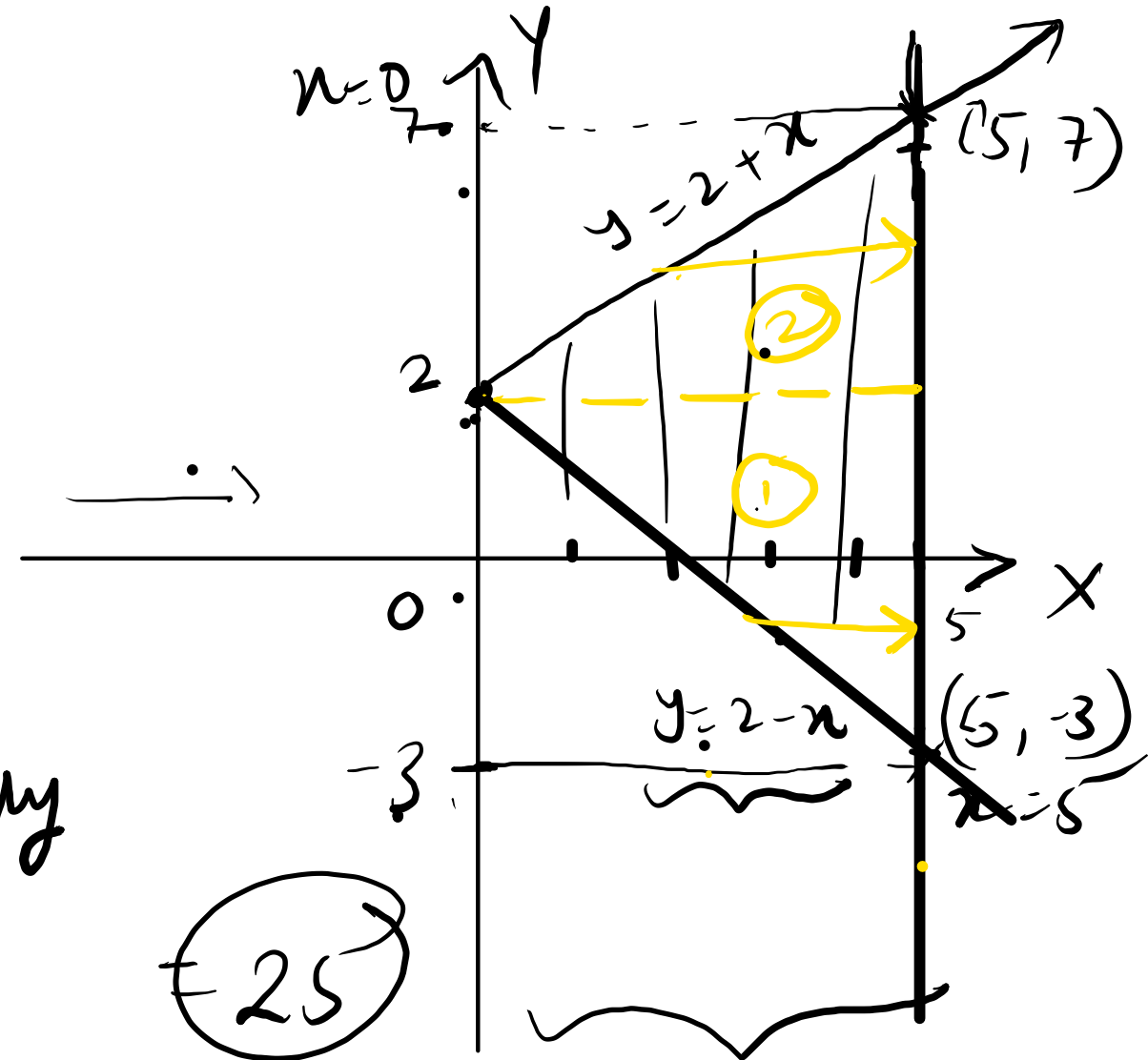
$$y=2+x$$

x	0	5
y	2	-3

x	0	5
y	2	7

$$\int_{y=-3}^2 \int_{x=2-y}^5 dx dy$$

$$+ \int_{y=-2}^7 \int_{x=y-2}^5 dx dy$$



17. $\int_0^a \int_{x^2/a}^{2a-x} xy \, dy \, dx$
 $x = y =$

$x=0 - x=a$
 $y = x^2/a \quad y = 2a-x$

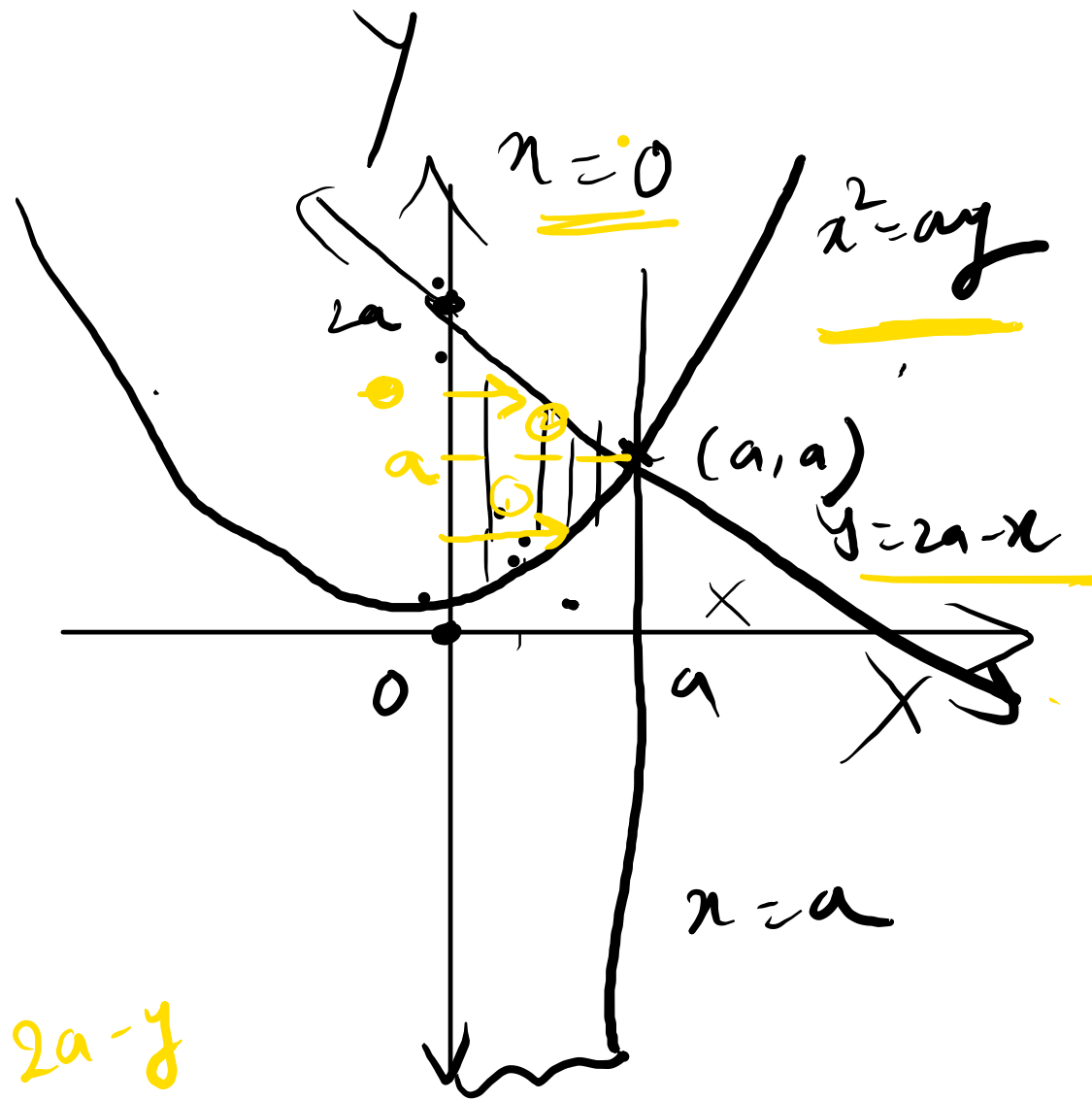
$x^2 = ay$

x	0	a
y	0	a

$x = \pm \sqrt{ay}$

put $x=a$
 $a^2 = ay$

$\int_{y=0}^a \int_{x=0}^{\sqrt{ay}} xy \, dx \, dy + \int_{y=a}^{2a} \int_{x=0}^{2a-y} xy \, dx \, dy = \frac{3}{8} a^4$





$$\int_{-1}^1 \int_{-\sqrt{y}}^{\sqrt{y}} dx dy$$

$$\int_0^1 \int_{-\sqrt{y}}^{\sqrt{y}} dx dy + \int_1^3 dy \int_{-1}^1 dx$$

$$y=0 \quad y=1$$

$$y=3$$

$$x=-1$$

$$x=1$$

$$x = \pm \sqrt{y}$$

$$x^2 = y$$

$$y=1$$

$$x = \pm 1$$

$$\int_{x=-1}^1 \int_{y=x^2}^3 dy dx$$

$$dy dx = \frac{16}{3}$$

